A

Shared and distinct expression signatures

Union of top 8 author-FDR genes per condition

Ddx3y
Uty
Kdm5d
Xbp1
Zbtb40
Creld2
Lyrm9
Hunk
Pcsk6
Abca8a
Tmem114
Kirrel3
Igkc
ENSMUSG00000092492
Sdk1
ENSMUSG00000056032
Nr1d1
Nlgn1

PM2.5
PFF
PM–PFF
PM–PFF + PM2.5

Shrunken log₂FC vs PBS

3
2
1
0
−1
−2
−3

B

Recomputed four-condition intersections

Genes in exact intersection

600
500
400
300
200
100
0

582
444
323
211
161
156
49
33
26
18
7
6
3

PM2.5
PFF
PM–PFF
PM–PFF + PM2.5

C

Author immune leading edge · model uncertainty

H2-Aa
H2-Eb1
H2-Ab1
Cd74
Il4ra
Igha
Crhr1
Stx4a

0
1
2
3
4
5

PM2.5 vs PBS MLE log₂FC ±1.96 SE